

Professional development

Table 1. Skills matrix

Skill area	Before this module	After this module	Evidence
Research question design	I could state a topic interest, but my questions were broad and not always linked to measurable outcomes.	I frame focused research questions that map to aims, objectives, and evaluation, with clear OT constraints in mind.	Research Proposal Presentation Transcript. Slide 4 (RQs) + Slide 3 (aim/objectives).
Literature review planning	I searched for sources, but I did not use consistent inclusion criteria or a clear synthesis framework.	I use databases, keywords, and time bounds, then organise findings with a synthesis framework tied to threats and controls.	Literature Review Outlines + Guide Questions – Answers.
Method justification	I tended to pick methods by familiarity, with limited explanation of why they fit the problem context.	I justify a staged design (Design Science + targeted qualitative feasibility input) aligned to deliverables and evaluation needs.	Research Proposal Presentation Transcript. Slide eight: Research Proposal Review.
Quantitative analysis	I understood descriptive statistics, but I lacked confidence selecting and interpreting hypothesis tests.	I can state hypotheses, choose suitable tests (paired/independent/one tailed), interpret p-values, and report decisions clearly.	Hypothesis Testing Worksheet – Research Version: Summary Measures Worksheet – Research Version.
Research writing	My writing was sometimes descriptive, and I needed stronger signposting and tighter argument flow.	I write in structured sections (problem–method–ethics–evaluation) and support claims with consistent Harvard citations.	Research Proposal Presentation Transcript + reflective writing samples.
Ethics in computing	I treated ethics as a checklist and focused on technical controls rather than governance and user impact.	I critically consider consent, transparency, accountability, and professional duties alongside security measures.	Reflective Activity A Emails: Inappropriate Use of Surveys case analysis.

Table 2. SWOT and action plan

Area	Point	Action I will take	Timeline
Strength	Strong OT/SCADA context awareness and ability to translate security ideas into feasible controls.	Use feasibility criteria (uptime, patch windows, legacy limits) to prioritise controls and justify decisions.	Next proposal iteration (Week 1–2)
Weakness	Evaluation metrics and measurement plan can be tightened to avoid vague conclusions.	Define 4–6 metrics (e.g., latency impact, detection quality, false positives, feasibility score) and link each to a research question.	Within 1 week
Opportunity	Security toolkit can become a reusable professional artefact aligned to NIST/IEC guidance.	Package templates, checklists, and testbed steps as a versioned toolkit with a short user guide and evidence examples.	2–3 weeks
Threat	Limited access to practitioners and the risk of scope creep or testbed complexity affecting timelines.	Keep interviews optional; rely on simulated data: apply change control and a small risk register to protect scope.	Ongoing (review weekly)

Professional Development Plan (PDP)

Objective	Key actions	Evidence / success criteria	Support / resources	Target date
Strengthen evaluation design for the research proposal	Define 4-6 evaluation metrics (e.g., feasibility score, latency impact, false positives); link each metric to a decision in the artefact; document assumptions.	Updated proposal evaluation section: metric table included clear justification in method.	NIST SP 800-82; IEC 62351; module notes; supervisor feedback if available.	Week 1
Improve Harvard referencing consistency and academic writing flow	Standardise in text citations; run a reference checklist; ensure every figure/table is captioned and referenced in text.	No missing references; consistent formatting; improved readability in unit summaries and evaluations.	Library guidance; UoEO study skills hub; reference manager (optional).	Week 2
Develop stronger quantitative reporting habits	Complete two extra practice problems (hypothesis testing + effect size); write short method result interpretation notes.	Two worked examples in appendix; assumptions stated, interpretation avoids confusing statistical and practical significance.	Units 8-9 worksheets; lecture notes; practice datasets.	Week 2-3
Create an ethics and governance checklist for AI/OT use cases	Draft checklist covering transparency, access control, logging, data minimisation, retention, accountability; apply it to the proposal artefact.	Checklist included as an appendix; proposal ethics section updated with concrete controls and responsibilities.	ACM/BCS guidance; GDPR principles; module materials.	Week 3

Table 3. Professional Development Plan (PDP).